

# Zhihan Rachel Hua

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## SUMMARY

Ph.D.-trained computational biologist with extensive experience developing AI-enabled bioinformatics workflows for genomics and multi-omics research. Specialized in designing scalable analysis pipelines for spatial transcriptomics, single-cell sequencing, long-read genomics, and epigenomics, with a strong emphasis on reproducibility, automation, and biological interpretation. Experienced working across research and product teams to build computational tools that accelerate scientific discovery and support translational research.

## SKILLS

**Programming & Data Science:** Python, R, SQL, Bash, Unix/Linux, Shell Scripting, Pandas, NumPy, SciPy, scikit-learn, Jupyter, VS Code

**Single-cell & Spatial Genomics:** Scanpy, Seurat, Squidpy, AnnData, CellTypist, spatial transcriptomics (10x Xenium, Visium, CODEX/PhenoCycler), spatial statistics, neighborhood/niche analysis, cell-type annotation, ligand-receptor analysis, multi-omic integration (Harmony, MaxFuse, uniPort)

**Genomics & Bioinformatics:** RNA-seq (DESeq2, edgeR, limma), single-cell RNA-seq, pseudobulk analysis, pathway enrichment (GSEA, fgsea, CAMERA), long-read sequencing (Oxford Nanopore, PacBio), variant calling (GATK, BCFtools, samtools), alignment (BWA, Bowtie2, minimap2), GWAS (PLINK), genome assembly & QC (SPAdes, QUAST), variant annotation (snpEff), epigenomics (CUT&Tag, MACS3, deepTools), visualization (IGV)

**Machine Learning & AI:** AI-assisted bioinformatics workflows, LLM applications for scientific analysis, retrieval-augmented generation (RAG), statistical modeling, clustering, dimensionality reduction (PCA, UMAP, t-SNE), model benchmarking

**Workflow Automation & Infrastructure:** Nextflow, Snakemake, Docker, HPC (SLURM), AWS, Google Cloud Platform (GCP), Lambda Cloud, Vercel, GitHub Actions, pipeline development, cloud deployment, reproducible workflows

**Version Control & Software Engineering:** Git, GitHub, CI/CD, Docker, code review, software documentation, API integration

## RESEARCH EXPERIENCE

**Bioinformatics Engineer** April 2026 – Present

*Novaflow (YC-backed bioinformatics platform) | Los Angeles, CA*

- Lead bioinformatics analyses across 10+ client projects spanning academia, biotechnology, and pharmaceutical research, delivering interactive web portals with reproducible workflows, downloadable datasets, publication-quality figures, and technical reports.
- Analyze multi-omics datasets including spatial transcriptomics (10x Xenium, CODEX/PhenoCycler), single-cell RNA-seq, long-read sequencing (Oxford Nanopore), epigenomics (CUT&Tag), and translational profiling (Ribo-seq/Polysome-seq) to generate biologically meaningful insights.
- Apply modern protein structure prediction and design tools, including AlphaFold, ESMFold, and RFdiffusion, to model protein structures, assess model confidence, evaluate protein-protein and protein-ligand interactions, estimate binding affinity, and prioritize variants based on predicted stability and function.
- Performed structure-guided antibody optimization targeting Ephrin-B2 and EphB4, using computational protein modeling and in silico mutagenesis to evaluate candidate mutations, predict changes in binding affinity and structural stability, and prioritize variants for improved target binding.
- Develop end-to-end analysis pipelines for preprocessing, quality control, clustering, cell type annotation, differential expression, pathway enrichment, spatial analysis, and integrative multi-omics workflows.
- Design and evaluate computational methods for integrating diverse omics modalities, benchmarking state-of-the-art algorithms and selecting appropriate analytical strategies based on study design and data quality.
- Perform quality assessment and troubleshooting of high-throughput sequencing experiments, identifying technical artifacts and implementing reproducible analysis pipelines for downstream processing.
- Reanalyze published and proprietary genomic datasets using appropriate statistical frameworks, improving analytical rigor through robust experimental design and correction of batch effects, sequencing depth, and pseudoreplication.
- Build and maintain scalable cloud computing infrastructure across AWS GPU/HPC environments, supporting large-scale genomics analyses and deployment of interactive bioinformatics applications.
- Develop internal bioinformatics tools and AI-enabled workflows to improve data processing, literature ingestion, variant interpretation, and automation of genomic analyses.
- Prepare technical reports, scientific presentations, and client-facing deliverables, translating complex computational analyses into clear biological insights for academic and industry collaborators.
- Mentor junior bioinformaticians and interns by reviewing code, analysis pipelines, and experimental designs, while contributing to internal method development and platform improvements.

**Postdoctoral Fellow** Sept 2025 – April 2026

*Ambry Genetics Tempus AI | Aliso Viejo, CA*

- Executed and optimized PacBio long-read sequencing workflows, including sample selection, DNA extraction, library

preparation, and QC, enabling consistent, high-quality data generation on a weekly basis and driving the development of a scalable, productized sequencing pipeline.

- Partnered with cross-functional stakeholders (scientists, automation engineers, operations) to gather requirements and co-develop solutions, aligning technical workflows with business and clinical objectives, including designing and optimizing liquid-handling workflows on Tecan and Hamilton systems for scalable, high-throughput sequencing operations.
- Performed and built downstream NGS bioinformatics pipelines to identify structural variants and genome-wide features from long-read datasets.
- Applied American College of Medical Genetics and Genomics (ACMG) criteria to interpret and curate variants from PacBio long-read sequencing data with a focus on rare disease research, delivering clinically actionable insights and presenting results in weekly team meetings.
- Contributed to process documentation, workflow standardization, and system optimization, supporting reproducibility, compliance, and knowledge transfer across teams.
- Evaluated and recommended improvements to scientific data workflows, enhancing data integration, traceability, and usability across systems.

#### **Graduate Student Researcher** *August 2020 – August 2025*

*Nuzhdin Lab at USC | Los Angeles, CA*

- **Transcriptomic Analysis of Organism Development:** Spearheaded RNA expression analysis in 3-day-old and 7-day-old larvae, uncovering critical biological pathways and deformity-associated genes. Developed and optimized customized R/Python scripts for transcriptomic and functional genomics analysis, applying statistical methods to interpret large-scale datasets using HPC and cloud computing (AWS, GCP).
- **Germline Mutation Detection:** Designed, built, and optimized high-throughput WGS bioinformatics workflows and custom Python/R/Bash scripts to detect de novo germline mutations in bivalves. Incorporated advanced variant calling, joint genotyping, and tailored QC filters (depth, allelic balance, strand bias) to reduce false positives, enabling robust mutation-rate estimation and advancing understanding of genetic variation in marine invertebrates.
- **Machine Learning & Biomarker Discovery:** Designed and executed a field-based molecular ecology project on wildfire-related stressors in marine ecosystems. Developed predictive models to identify key molecular biomarkers and differentially expressed genes associated with stress response and immune function, integrating RNA-seq with environmental metadata to reveal enriched pathways and population-level effects.
- **Genomic Selection and Trait Mapping:** Applied genomic-assisted breeding strategies to bivalve aquaculture by analyzing multi-omics data and identifying SNPs linked to growth, survival, and stress resilience. Conducted GWAS (mixed linear models) and selective-sweep analyses (XP-CLR, FST) to detect trait-associated loci and regions under selection, delivering actionable breeding recommendations.
- **Mutagenesis Variant Calling Study:** Designed a forward-genetic-screening pipeline for *Macrocystis pyrifera* gametophytes using EMS-induced point mutations, immobilizing mutant cells for phenotypic screening and performing variant calling to map causative mutations linked to motility defects and developmental disruption.

#### **Biomedical Engineer Intern** *June 2018 – September 2018* *VoxelCloud | Los Angeles, CA*

- Developed and maintained a deep-learning AI pipeline to enhance diagnostic accuracy of medical imaging, achieving a 30% improvement in diagnostic performance.
- Analyzed 200+ medical images, ensuring a 95% precision rate for pipeline diagnostics and validating clinical reliability. • Delivered documentation and quality reports and collaborated with cross-functional teams to integrate research findings into the product development pipeline.

## **EDUCATION**

**University of Southern California — Ph.D., Computational and Molecular Biology** 2020 – 2025 **University of California, Los Angeles — B.S. in Biochemistry** 2016 – 2020

## **PUBLICATIONS AND PRESENTATIONS**

1. Zhihan Hua, Nathan Churches, Sergey V. Nuzhdin. Balancing selection and candidate loci for survival and growth during larval development in the Mediterranean mussel, *Mytilus galloprovincialis*. *G3 Genes|Genomes|Genetics*, 13(7), July 2023, jkad103.
2. Zhihan Hua, Nathan D. Churches, Jordan Chancellor, Peter L. Chang, Sergey V. Nuzhdin. Pacific oysters (*Crassostrea gigas*) dramatically recalibrate models for the upper limit of the eukaryotic mutation rate (manuscript in review).
3. Zhihan Hua, Sergey Nuzhdin. Population Genomics of Mediterranean Mussels, *Mytilus galloprovincialis*, Uncovers Divergent Selection Signatures Between Core and PAV Gene Families (manuscript in review).
4. Zhihan Hua. A corrected ranking axis for spatial-transcriptomics translation: report  $z(\text{PCC}) + z(\text{LISA})$ , because PCC-only leaderboards misrank methods for downstream spatial analysis (manuscript in review).